The activation function decides what is solvable.

Know which one and the task is easy. When you don't, **evolution can find it, one function per node.**

Per-Node Activation Function Evolution in Indirectly Encoded Substrates: Solvability, Limits, and Emergent Diversity

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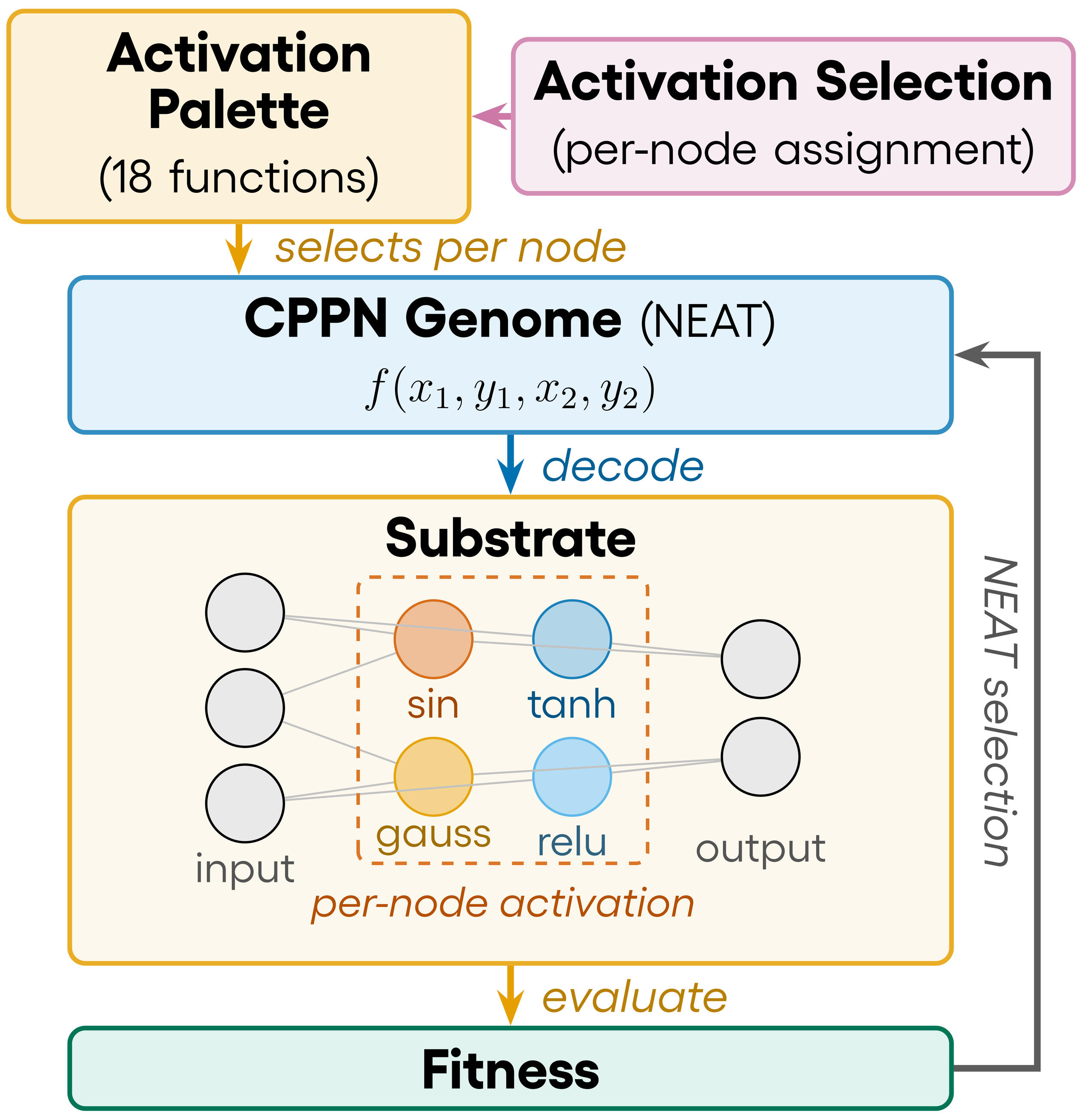
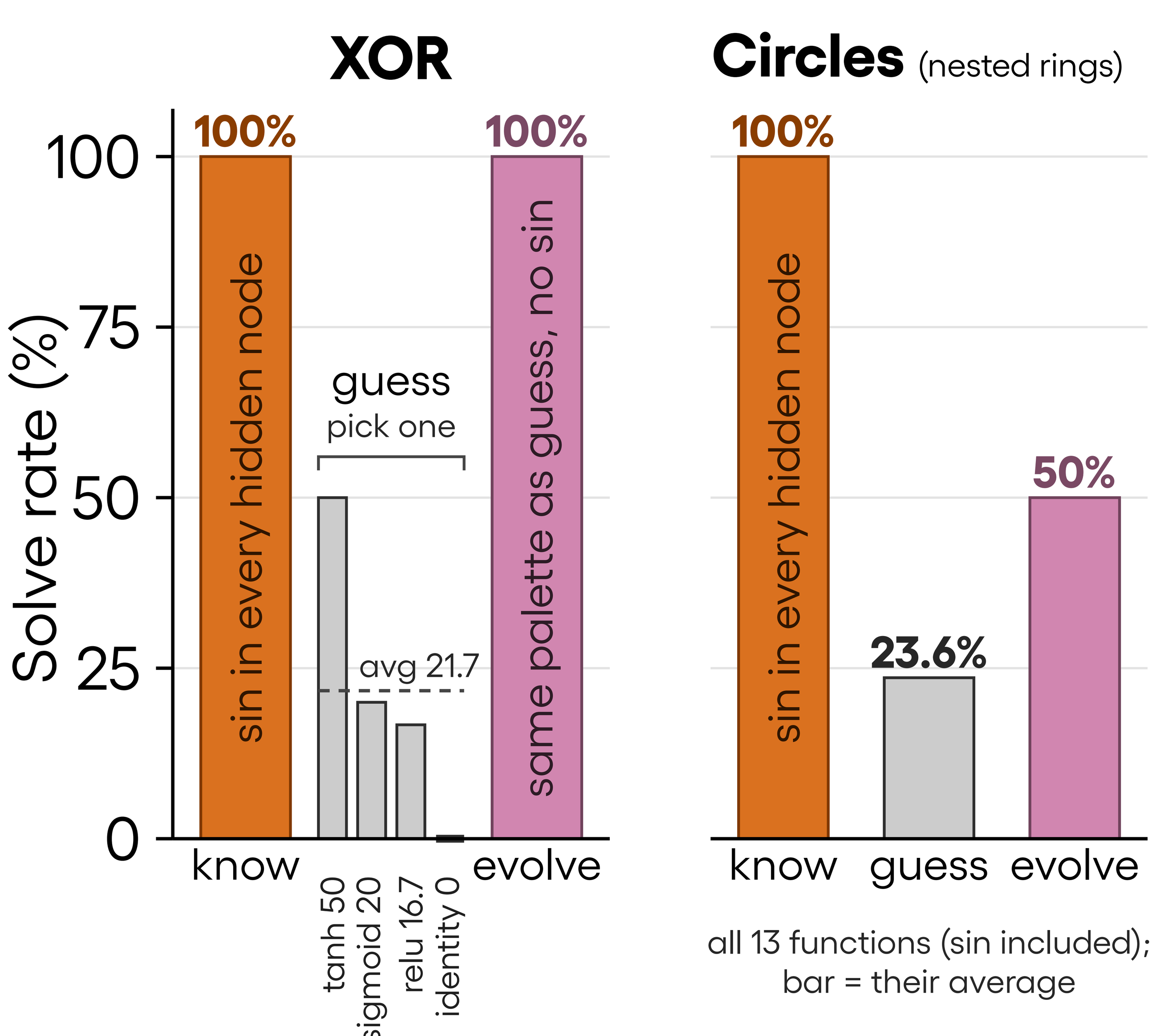
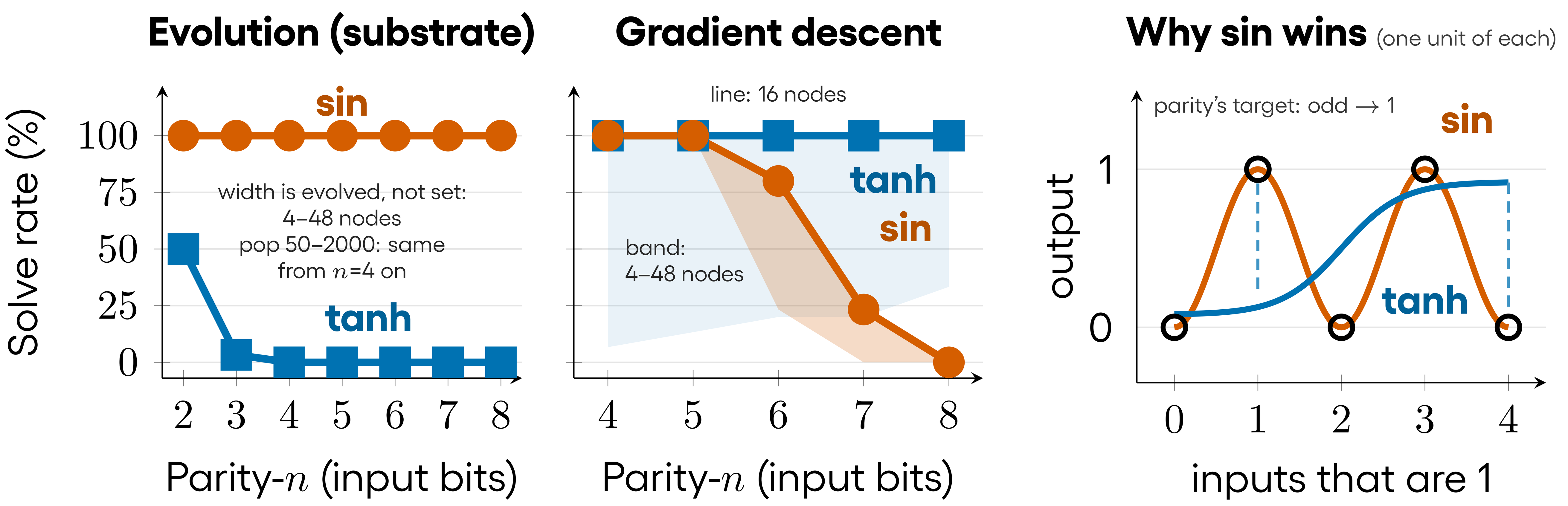


Know it, guess it, or evolve it?



Know: sin, a right answer, in every hidden node. Guess: one function for all nodes. Evolve: the CPPN picks per node. ***XOR's palette: tanh, sigmoid, relu, identity. No sin, yet evolution matches sin anyway.*** *Circles has all 13 with sin; that freedom costs 50% vs 100%. 30 replications each.*



Why you often don't know: *the right choice inverts with the optimizer (30 replications each).*
Why sin wins: *parity alternates; a turning curve follows, a flat one cannot (dashed: misses).*

The problem

Brains mix cell types. Networks give every node the *same* activation, and that choice decides what the network **can** represent: on Parity-4, 4 of 18 functions always solve and 9 never do. More turns in a function's curve, more often it solves (ρ=0.81).

The approach

An **extra CPPN output channel** gives every node its own function from an 18-function palette; panels use 4- and 13-function subsets. 4,500+ runs: Boolean, regression, and spatial tasks, all $\leq$8 inputs.

Findings

Know it, it's trivial: sin solves Parity-4 in **1.7 gens**.

Guess it, it fails: the defaults tanh, relu, sigmoid score **0%** on Parity-4.

Evolve it, it works: a palette without sin still solves XOR **30/30**, at 16× sin's gens.

Why open-ended?

Nobody picks the functions: evolution assigns them node by node, building networks no designer would create.

Where it stops

Evolution cannot find what the palette lacks: XOR's four flat defaults score **0%** on circles. Gradient descent needs width: **2/30** to **30/30**.

TAKE-HOME

When you don't know, don't pick a function. Give evolution a palette.

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Paper & code

github.com/RomainClaret/emr-hyperneat